

How the Primes Work, in Five Pictures

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June 2026

Abstract

The ledger paper in this repository reports that ten prime-distribution phenomena pass numerical gates; it does not explain *why*. This companion supplies the why, at the level where the why is actually known: in five pictures. (1) Patterns of primes live or die cell-by-cell on each prime’s *wheel*, and the surviving fractions multiply into every density law Hardy and Littlewood wrote down. (2) That one multiplication predicts twin counts to 0.003%, the doubled abundance of gap-6 pairs, and Goldbach counts per N . (3) Read through the explicit formula, the primes are a sum of waves whose spectrum peaks exactly at the zeros of the zeta function — we plot this, and the peaks land on independently computed zeros. (4) A one-dimensional golden-ratio quasicrystal shows the same signature — sharp spectrum, no period — which is precisely Dyson’s reason for suggesting that the Riemann Hypothesis is secretly a statement about quasicrystals; this is where the question becomes geometric. (5) Everything open in the subject is one wall: the correlations of those zeros. All mathematics here is classical and attributed; the pictures and their code are ours. Nothing below proves, or claims to prove, any open conjecture.

1 The wheels: where the densities come from

Look at the whole number line through a wheel with three cells — the residues 0, 1, 2 modulo 3. Every integer lands on one cell; primes (other than 3 itself) only ever land on cells 1 and 2, because cell 0 means “divisible by 3.”

Now place a *pattern* on the wheel instead of a single number: the twin pattern $(n, n+2)$. If n is on cell 0, the pattern dies (n is divisible by 3). If n is on cell 1, then $n+2$ is on cell 0: dead again. Only n on cell 2 survives. **One cell of three.**

Compare with what independence would give: two unrelated numbers each avoid cell 0 with probability $2/3$, so independent survival would be $(2/3)^2 = 4/9$. The twin pattern’s actual survival is $1/3 = 3/9$. The wheel taxes twins by a factor

$$\frac{1/3}{(2/3)^2} = \frac{3}{4} = \frac{p(p-2)}{(p-1)^2} \Big|_{p=3},$$

and the same computation on every prime wheel $p > 2$ gives the factor $p(p-2)/(p-1)^2$: the pattern occupies two cells, so it dies on two of p , surviving $p-2$, against an independence

*Companion to *The Prime Field Through One Instrument* (the verification ledger) in this repository: github.com/vfd-org/closure-positivity-lab. That paper verifies; this one explains. Every figure here is computed, not drawn — the generating script is `figures/make-pictures.py`, and two of the pictures are interactive at `explorables/`.

baseline of $((p-1)/p)^2$. Multiply over all odd wheels:

$$\prod_{p>2} \frac{p(p-2)}{(p-1)^2} = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) = C_2 = 0.6601618\dots$$

This is the twin-prime constant of Hardy and Littlewood [1], and now you know what it *is*: the total tax the wheels charge the twin pattern, relative to independence. The predicted count below x is $2C_2 \int_2^x dt/\log^2 t$, and the sieve agrees to 0.003% at $x = 5 \times 10^7$ (ledger row 1).

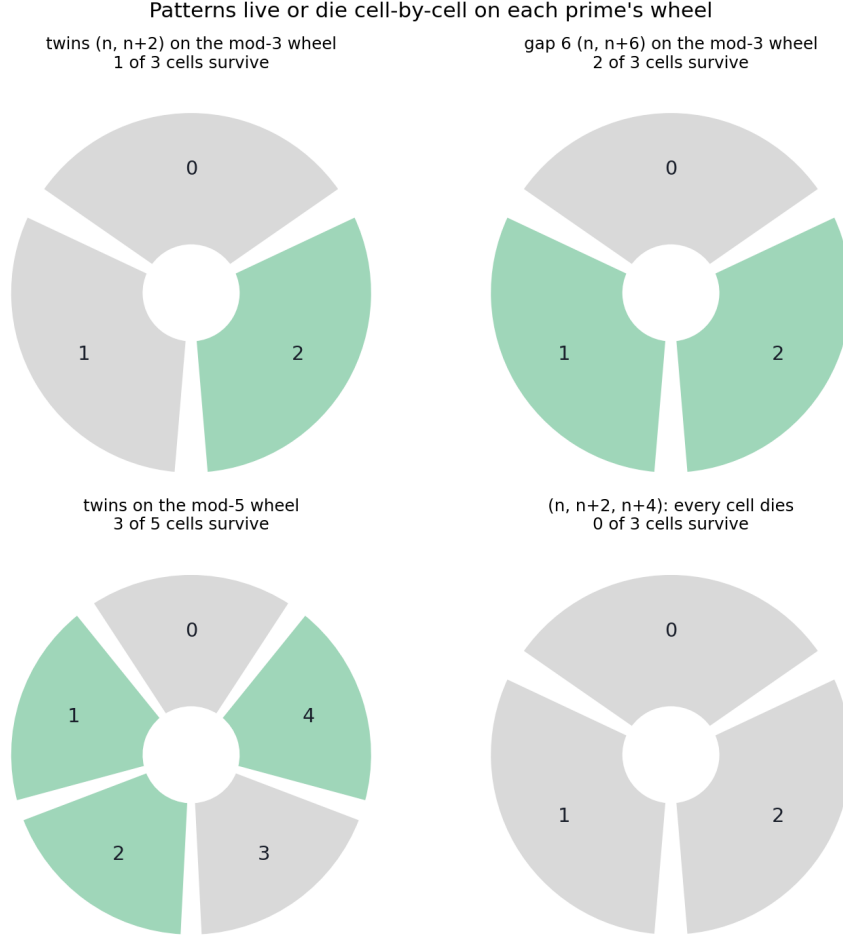


Figure 1: The mod-3 and mod-5 wheels. Green cells: starting positions where the pattern avoids every forbidden cell. Twins survive 1 cell of 3 (top left); the gap-6 pattern survives 2 of 3 (top right) — that single picture is the whole reason gap-6 pairs are twice as common as twins. Bottom right: $(n, n+2, n+4)$ kills every cell mod 3, so only one such triple exists in all of mathematics — $(3, 5, 7)$, where the dead cell is the prime 3 itself.

Two more readings of the same picture. **Why gap 6 doubles:** the pattern $(n, n+6)$ doesn't move on the wheel of 3 (adding 6 is adding 0), so it dies only on cell 0: two cells of three survive — *twice* the twin survival, and no other wheel distinguishes the two patterns. Hence gap-6 pairs are exactly twice as abundant; the sieve says 1.9950. **Why $(n, n+2, n+4)$ is**

impossible: on the wheel of 3 the three offsets cover all three cells, so every starting position dies. The only escape is the pattern that *includes* 3 itself: (3, 5, 7), once, never again. The ledger checks the analogous exact fact on the wheel of 5: the only twin pair $(p, p+2)$ with $p \equiv 3 \pmod{5}$ is (3, 5), because cell 3 forces $5 \mid p + 2$.

In the programme's language each wheel is a *closure condition at the place p*: a finite cyclic screen that a pattern either clears or does not. The density of any pattern is its product of survival ratios over all screens — an *all-finite-places product*, the same shape as the total-positivity law studied elsewhere in this repository. That parallel is structural, not decorative: in both cases the decidable content lives place-by-place, and the open content lives in how infinitely many places conspire.

2 One multiplication, every pattern

The wheel computation generalises mechanically: for a pattern of k offsets, count surviving cells $w(p)$ on each wheel, divide by the independence baseline, multiply. The result (the *singular series*) predicts, with *no further input*:

- the relative abundance of every even gap (figure below: 105 gaps, predictions as dots, sieve counts as bars — worst disagreement 0.25%);
- Goldbach counts per individual N — the wheels of the primes *dividing* N relax their conditions, so N divisible by 3 has about twice the representations of its neighbours (verified: 200 consecutive even N near 10^6 , mean deviation 0.65%);
- the twin count itself, to 0.003%.

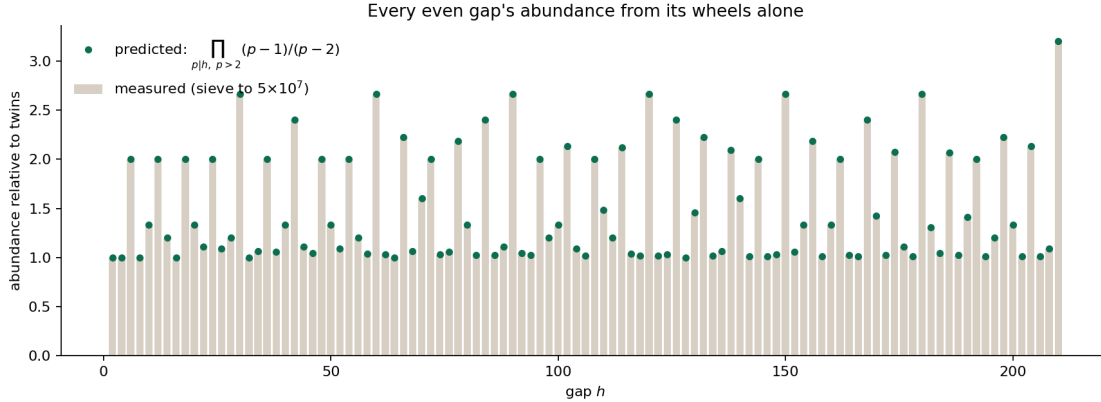


Figure 2: All even gaps to 210: measured abundance (bars) against the wheel product (dots). The spikes are gaps divisible by 3, 15, 105... — more surviving cells, more pairs. Interactive version: explorables/gap-slider.html.

What the wheels do *not* give is infinitude: they say twin pairs keep their density *if anything like statistical regularity holds*, but cannot rule out the counts simply stopping. That question changes character entirely, and needs the second half of the story.

3 The duality: primes are waves, zeros are the spectrum

Riemann's explicit formula is an exact identity, and it has a physical reading. Place every prime p on a logarithmic line and let it broadcast a wave of frequency $\log p$ (with weight $\log p/\sqrt{p}$, and its powers chiming in quietly). Sum the waves and take the spectrum — the same operation a crystallographer performs when she shines X-rays through a crystal and reads the diffraction pattern.

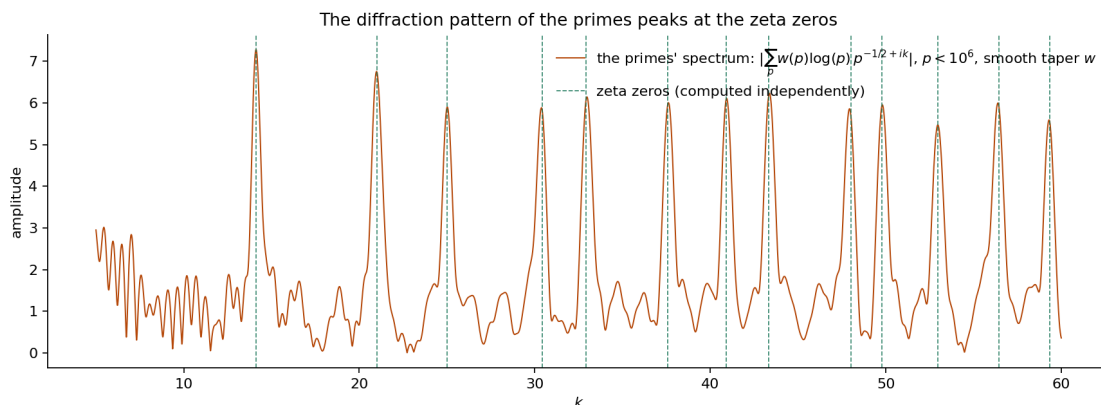


Figure 3: The spectrum of the prime waves (orange), computed from the primes below 10^6 with a smooth taper, and *nothing else*; dashed lines are the first zeta zeros, computed independently in PARI/GP from the zeta function. The peaks land on the zeros: 13 of the first 15 align within 0.15 (the misses are close zero-pairs merged at this resolution).

The picture says what the explicit formula says: **the zeros of ζ are the spectrum of the primes**. The identity runs both ways — zeros reconstruct prime counts, primes reconstruct zeros (the companion paper's prime-wave reconstruction of the critical line is this same duality run in reverse for our icosian L -function). Every question the wheels cannot answer is a question about this spectrum: the *error* in the prime count is a sum over zeros; the Chebyshev bias is an interference effect (the prime-*square* wave loads one residue class); twin-prime infinitude is governed by correlations *between* zeros. The wheels are the local physics; the spectrum is the global physics; the explicit formula is the exchange rate.

4 The crystal question: where it becomes geometry

A crystal produces sharp diffraction peaks because it is periodic. In 1982 Shechtman saw sharp peaks with a symmetry no period allows; the resolution is the *quasicrystal*, and its one-dimensional archetype is the Fibonacci chain — two tile lengths in ratio φ (the golden ratio), arranged by the substitution $A \rightarrow AB, B \rightarrow A$, never repeating.

Now hold figure 3 next to figure 4's top panel. The primes (suitably weighted, on the log line) also show a sharp spectrum without any period — *provided the peaks all sit where we plotted them*, namely on one vertical line in the complex plane. That proviso is the Riemann Hypothesis. This observation is due to Dyson [2], who proposed classifying one-dimensional quasicrystals as a route to RH: if the primes' spectrum is genuinely point-like, the primes are (in this precise sense) a quasicrystal, and quasicrystals might be classifiable.

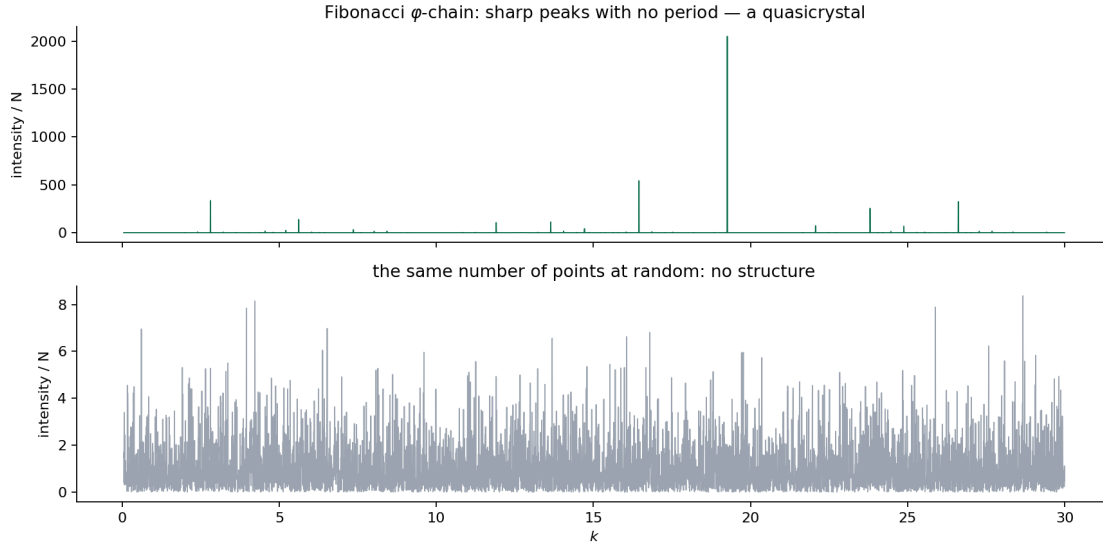


Figure 4: Computed diffraction of 3,000 points. Top: the Fibonacci φ -chain — razor-sharp peaks, no period. Bottom: the same number of points placed at random — featureless. Sharp spectrum without periodicity is possible, and golden-ratio geometry is its canonical home.

We state plainly what this is and is not. It is the one place where the deepest open question about primes becomes a question of geometry — the geometry of aperiodic order, whose golden-ratio examples (Fibonacci chains, Penrose tilings, the icosians of this repository) are the native objects of our wider programme. It is *not* a result: nobody has the classification, we do not claim it, and the honest status of every approach we have tested is recorded in the companion papers — including the negative results.

5 The wall, and the map

Assemble the picture. Everything decidable about prime patterns we have met lives *locally*: wheels, survival counts, one multiplication — verified across ten phenomena in the ledger at the 0.003%–0.25% level, with three theorem-grade anchor rows calibrating the instrument. Everything open lives *globally*, and it is one object, not many: the fine structure of the zero spectrum. Twin primes, Polignac, Goldbach, Cramér’s gap bound, the unconditional Chebyshev density — each is a different question about the same correlations (ledger row 10 measures them directly: spacing statistics of 1,983 computed zeros match the random-matrix law within the documented finite-height corrections, with level repulsion suppressed $70\times$ relative to Poisson).

	the local layer (wheels)	the global layer (spectrum)
what it fixes	whether a pattern occurs; its density	infinitude; error terms; extremes
status	decidable, computed, verified	open in every conjectural case
the “how”	counting cells on finite screens	unknown; best picture: diffraction
where geometry enters	each place is a finite cyclic screen	Dyson’s quasicrystal question

That is the full picture of the primes as honestly as we can paint it: a local mechanism

that is completely understood and astonishingly accurate, a global mechanism that is one open question wearing many costumes, an exact dictionary between them, and a geometric reading of the open half that remains a question. The ledger paper carries the gates, artifacts and falsifiers behind every number quoted here; the explorables let you drive the two halves yourself.

References

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- [9] VFD Programme, *Closure diffraction and the icosian L -function*, this repository (2026).